

Question				Marking details	Marks Available				
					AO1	AO2	AO3	Total	Maths * Prac **
4	(a)			18 = 3 marks 18.18/18.2 = 2 marks Substitution into equation = 1 mark $\frac{240}{3.3 \times (\log 10^4 / \log 10^1)} =$ Allow 24 for 3 marks 24.2 = 2 marks Substitution into equation = 1 mark $\frac{240}{3.3 \times (\log 10^4 / \log 10^1)} =$ 3 = 2 marks (use of hours instead of minutes) 3.3 = 1 mark (use of hours instead of minutes and not 1 dp)		3		3	3
	(b)	(i)		death phase/ suitable description; (1) nutrient/glucose run out, build-up of toxic waste/toxins (1)	2			2	
		(ii)		Dead cells are included therefore there {are higher numbers/ numbers stay constant} (1) death phase cannot be identified (1)	1	1		2	
				Question 4 total	3	4		7	3